

## 3D Coordinates Extension — English Worksheet (Intermediate)

**Topic:** Design Your Own 3D Build (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Intermediate, after Lesson 3) · **Time:** about 45 minutes

Pixel art was flat — two numbers, one wall. This lesson your build is **solid**. Three numbers. You draw it from the **front** and from **above**, then read both drawings back into the world.

**Build your canvas.** Run this to lay a blank **15 × 15** floor, then put a **red block at your feet** as your **home spot**:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
               pos(1, 0, 1),
               pos(15, 0, 15),
               FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

Your build stands on that floor. **x** goes across, **y** goes up, **z** goes away from you — a **15 × 15 × 15** space.

● **Stand on the red block every time you run.** If you move, the build moves too.

### 1 · Predict 🧠

Read each set of steps. Write a sentence about what you will see.

```
place blue block at (3, 9, 4)
place blue block at (4, 9, 4)
place blue block at (5, 9, 4)
```

**Three blocks share y = 9 and z = 4. What shape do they make, and which number changes?**

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```
place blue block at (7, 2, 4)
place blue block at (7, 3, 4)
place blue block at (7, 4, 4)
```

**These three share  $x = 7$  and  $z = 4$ . What shape do they make, and which number changes?**

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```
place blue block at (7, 5, 1)
place blue block at (7, 5, 2)
place blue block at (7, 5, 3)
```

**These three share  $x = 7$  and  $y = 5$ . Which way does this line run, and which number changes?**

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```
place green block at (6, 6, 6)
place blue block at (6, 6, 6)
```

**Two blocks aim for the same  $(x, y, z)$ . What do you see — green, blue, or both? Why?**

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## 2 · Spot the Bug 🐛

Each one is broken. Fix it, then explain why the original was wrong.

**Bug A** — The table's two legs should sit at (2, 1, 2) and (2, 1, 4). Both land in the **same spot**.

```
place brown block at (2, 1, 2)
place brown block at (2, 1, 2)
```

**Write the fixed line:**

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**Why was it wrong?**

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**Bug B** — This should be a line running **away** from you at  $x = 8$ ,  $y = 5$ .

```
place black block at (8, 5, 3)
place black block at (9, 5, 3)
place black block at (10, 5, 3)
```

**Write the fixed lines:**

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**Why was it wrong? Why does your fix work?**

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### 3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

**Your five rules.**

| # | Your design must...                                                  |
|---|----------------------------------------------------------------------|
| 1 | fit inside the <b>15 × 15 × 15</b> space                             |
| 2 | use <b>3 or more</b> different block colours                         |
| 3 | use <b>40 or more</b> blocks                                         |
| 4 | be <b>recognisable</b> — someone else can name it without being told |
| 5 | be <b>drawn on both grids below before you place a single block</b>  |

**Two drawings, one build.** The front view gives you **x** and **y**. The top view gives you **x** and **z**. Together they place every block.

**Here is one I made** — a tree:

🌳 **Front view — x across, y up**

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   | ■ | ■ | ■ |   |
| 4 |   | ■ | ■ | ■ |   |
| 3 |   | ■ | ■ | ■ |   |
| 2 |   |   | ■ |   |   |
| 1 |   |   | ■ |   |   |

🌳 **Top view — x across, z away from you**

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 5 |   |   |   |   |   |
| 4 |   | ■ | ■ | ■ |   |
| 3 |   | ■ | ■ | ■ |   |
| 2 |   | ■ | ■ | ■ |   |
| 1 |   |   |   |   |   |

**Read it back.** The trunk is at  $x = 3, z = 3$ , and stands from  $y = 1$  to  $y = 2$ . The leaves fill  $x = 2$  to  $4$  and  $z = 2$  to  $4$ , at  $y = 3$  to  $y = 5$ .

**Now draw yours.** Colour the squares.



**Count as you go.** How many blocks did you use? How many colours?

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**Then build it.** Read the front view for **x** and **y**. Read the top view for **x** and **z**. Place each block at its (x, y, z).

**Check your work.** Walk around your build. Look at it from the front, then from above. Does each view match your grid? Write down any block you had to move.

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Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 • Your grids.** Hold this paper up to the camera.

**2 • Your build.** Point the camera. Hold your paper next to the build so both are in the shot.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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**Submit** 

Send this worksheet + your video to teacher on KakaoTalk.